TrES-3b

R-1183

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_TRES-3_220527` · created 2026-07-31T04:42:43+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459726.751 +/- 0.0017 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.046 +/- 0.057
Transit depth (Rp/Rs)^2	0.21 +/- 0.52 [%]
Orbital Inclination (inc)	77.99 +/- 2.87
Airmass coefficient 1 (a1)	2.06 +/- 1.61
Airmass coefficient 2 (a2)	-0.57 +/- 0.51
Scatter in the residuals of the lightcurve fit is	80.06 %
Best Comparison Star	None
Optimal Aperture	2.05
Optimal Annulus	3.0
Transit Duration (day)	0.008 +/- 0.017

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.046 ± 0.057 vs 0.16309 (deviation 2.05σ)
Duration fit vs published	0.008 d vs 0.0567 d (deviation 2.86σ)
Mid-time O–C	-4.2 min
Depth SNR	0.8
Residual scatter	80.06 %

Light curve

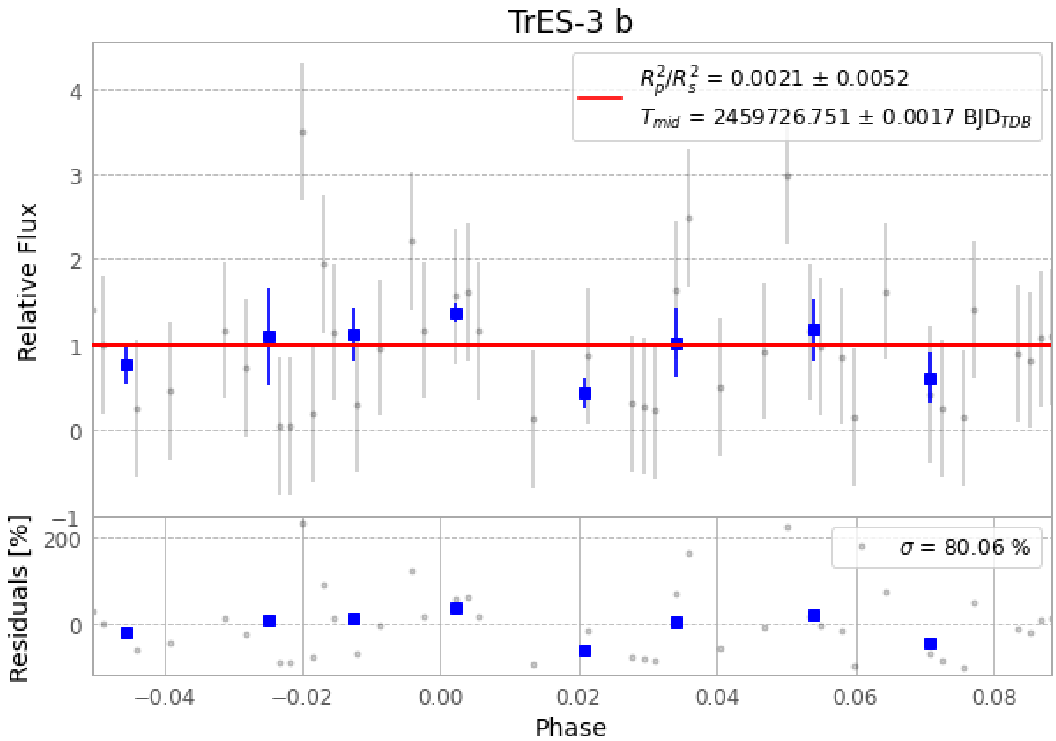

Figure 1 — FinalLightCurve_TrES-3 b_2022-05-26.png (SHA-256 8be0b7c053d9a56c…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [206, 242], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 9a6a57b0ce5a5544…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

88 FITS frames (58 MB), `TRES-3220527042113.FITS` → `TRES-3220527084213.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-220527.log (f3a34d97d69a…), FinalLightCurve_TrES-3 b_2022-05-26.png (8be0b7c053d9…), FinalLightCurve_TrES-3 b_2022-05-26.csv (965d3d8539a9…)

Compute resources

Wall time 120.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)